

AUTO SOLAR PILOT: Sunflower-Inspired Dual-Axis Solar Tracking System Using ESP32

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Abstract—This paper presents the design, development and implementation of an affordable sunflower-inspired dual-axis solar tracking system named *Auto Solar Pilot*. The system uses Light Dependent Resistors (LDRs), servo motors and an ESP32 microcontroller to continuously track the sun and align a photovoltaic panel for improved energy capture. The design mimics heliotropism observed in sunflowers and uses low-cost, widely available components to keep the system economical and easy to reproduce. Field tests demonstrate improved power output compared to fixed panels, and the architecture supports further IoT-enabled monitoring and enhancements.

Keywords:Solar Tracking, ESP32, LDR, Dual-Axis Tracker, Photovoltaic, Biomimicry.

I. BACKGROUND

The growing global need for sustainable energy has sped up the use of renewable technologies like solar power. Solar energy is a good alternative to fossil fuels because it is plentiful, clean, and renewable. But the effectiveness of solar photovoltaic (PV) panels depends a lot on how they are facing the sun. Fixed solar panels are easy to set up and don't cost much, but they don't stay in the best position to get sunlight all day, which wastes a lot of energy.

Solar tracking systems have been made to automatically change the angle of solar panels so they can follow the sun's path. This solves this problem. These systems make sure that sunlight hits the panel surface at a right angle, which makes power generation more efficient. Dual-axis trackers are the best type of tracker because they can adjust for both daily and seasonal changes in the sun's position.

The Auto Solar Pilot project is based on the natural phenomenon of heliotropism seen in sunflowers. It aims to create a dual-axis solar tracking system that mimics the way plants follow the sun. The system uses cheap Light Dependent Resistors (LDRs), servo motors, and an ESP32 microcontroller to automatically and accurately align the solar panel. This biomimetic method makes things more energy-efficient while keeping them cheap and easy to use.

In today's world, automation and IoT integration are very important for getting the most out of renewable energy systems. The ESP32 microcontroller has Wi-Fi built in, which means that it could be used for things like real-time data monitoring and performance analysis in the future. This means that the proposed system is not only a working prototype, but also a scalable and educational example of how to develop technology in a way that is good for the environment.

II. OBJECTIVES

The main goal of the *Auto Solar Pilot* project is to create and put into action a smart solar tracking system that is based on sunflowers and can collect the most solar energy with the least amount of resources. The exact goals are as follows:

- **To create a solar tracking system with two axes:** Make a system that follows the sun's movement both up and down and side to side so that it gets the most sunlight possible all day and all year.
- **To use parts that are cheap and easy to get:** Use LDR sensors, servo motors, and an ESP32 microcontroller to make a tracking system that works well and doesn't cost too much.
To make energy harvesting more efficient: By keeping the panels in the best position, you can get a measurable increase in power output compared to fixed solar panels.
- **To make sure it can work on its own:** Use a fully automated control algorithm that can make changes in real time without any help from a person.
- **To check how well the system works:** Do experimental tests in real sunlight to see how much energy is gained, how accurate the tracking is, and how reliable the whole system is.
- **To look into future scalability:** Give a base for combining IoT features and predictive control methods for better management of solar energy.

The project seeks to illustrate that intelligent, nature-inspired automation can greatly improve the utilization of renewable energy while ensuring economic viability and design simplicity.

III. METHODOLOGY

The methodology explains how to build, design, and use the solar tracking system that looks like a sunflower. It uses both hardware and software to automatically align solar panels based on real-time sunlight detection. The suggested system has four LDR sensors in the corners of a cross-like frame that is attached to the solar panel. These sensors send analog voltage signals to an ESP32, which then calculates the difference in light intensity between pairs of sensors that are opposite each other. According to the comparison, servo motors move the panel so that it faces the sun directly by turning it both horizontally and vertically.

A. Hardware Components and Setup

The first step involves selecting and integrating the necessary components for the solar tracker:



Fig. 1: ESP32 Microcontroller

1) *ESP32 Microcontroller*: The ESP32 microcontroller is the best choice for IoT apps that need to send data in real time because it has a lot of processing power, is cheap, has a dual-core processor, and comes with Wi-Fi built in.

2) *LDR (Light Dependent Resistor)*: A Light Dependent Resistor (LDR), also called a photoresistor, is a device that can tell how bright the light is around it. It is perfect for measuring light levels because its resistance goes down as the light level goes up. The LDR sends out an analog signal that the ESP32's analog-to-digital converter (ADC) can easily read.

3) *Servo Motor*: In solar tracking systems, a servo motor is a precise actuator that moves the solar panel. The microcontroller (like the ESP32) sends it control signals to turn the panel so that it gets the most sunlight. The servo motor allows for precise angular movement within a certain range, which keeps the panel in the best position to catch the sun all day. By maximizing solar exposure, this makes the solar tracking system work better and produce more energy.

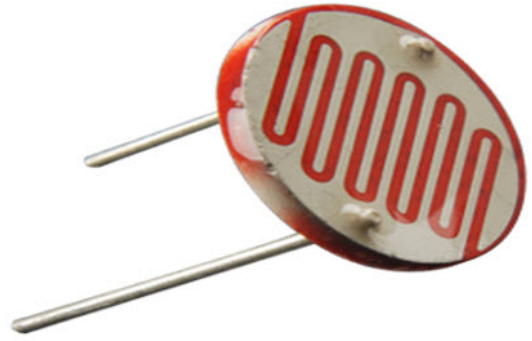


Fig. 2: Light Dependent Resistor

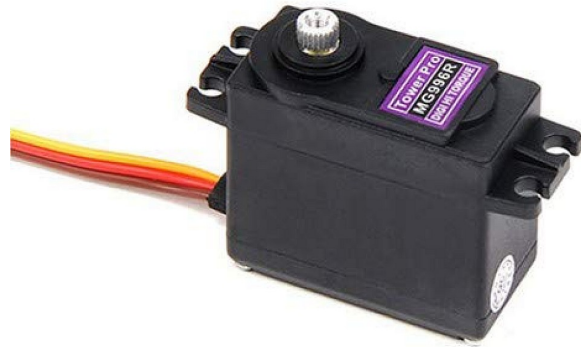


Fig. 3: Servo Motor

4) *Solar Panel*: Photovoltaic (PV) cells in a solar panel turn sunlight into electricity. It makes a DC voltage output that is proportional to the amount of sunlight, which makes it a renewable and environmentally friendly power source. The panel is the main source of power for the circuit in solar tracking or IoT-based energy systems. This makes sure that the system can run all day long.

B. Sensor Integration and Circuit Design

The ESP32 microcontroller is the main control unit in the solar tracking system. It connects to sensors and actuators to make sure the solar panel is always pointed at the sun to get the most energy. This is how the parts are connected:

1) *LDR Connection*: Four Light Dependent Resistors (LDRs) are placed in different corners of the solar panel to measure how bright the sun is. The ESP32's analog input pins get an analog signal from each LDR that shows how bright the light is. The difference in light levels can help you figure out where the sun is.

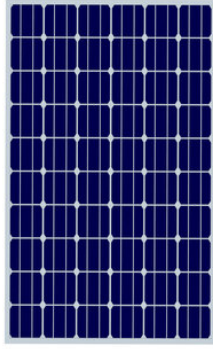
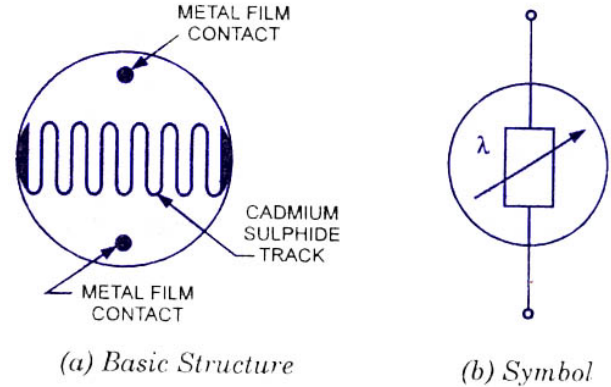


Fig. 4: Solar Panel



LDR

Fig. 6: LDR Circuit

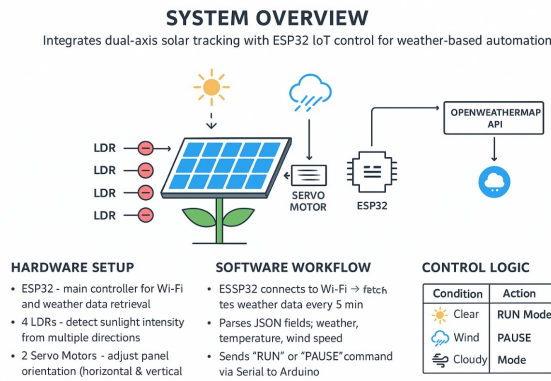


Fig. 5: Working Process

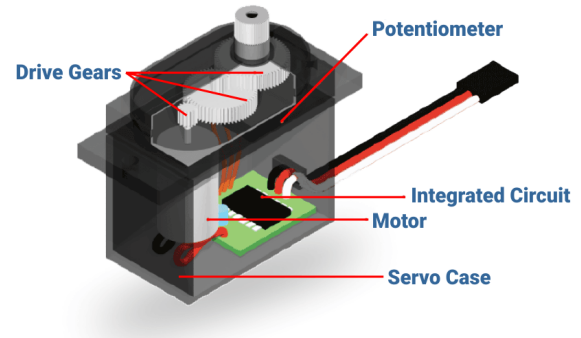


Fig. 7: Servo Motor Construction

2) *Servo Motor Connection:* The PWM output pins on the ESP32 are connected to the servo motors. The ESP32 sends control signals to the horizontal and vertical servo motors based on the LDR readings. This changes the position of the solar panel both horizontally and vertically so that it can follow the sun better.

3) *Power Supply:* A solar panel connected to a voltage regulator (LM7806) powers the system and gives the ESP32 and other parts a steady 6V DC output. There may also be a rechargeable battery in the power setup to store extra energy and provide power when it's dark or at night.

C. ESP32 Firmware and Programming

The Arduino IDE is used to program the ESP32 microcontroller. It has the right libraries and logic to read sensor data, process it, and control the servo motors so that the solar panels can follow the sun in real time. The firmware makes sure that the panels move smoothly and are always facing the sun.

1) *Library Import and Setup:* To control the servo motor and optionally send data remotely, you need to import the right libraries, like `Servo.h` and `WiFi.h` (if you use IoT monitoring). The ESP32 sets up the LDR sensors and connects the servo motors to the right PWM pins in the setup function.

2) *LDR Data Acquisition:* The ESP32 keeps reading the four LDR sensors on the solar panel's analog input values. Each LDR sends out a voltage signal that changes based on how bright the light is that hits it. These readings help us figure out which way has more light.

3) *Decision-Making and Control Logic:* A comparison algorithm looks at the LDR readings to find differences in light intensity between pairs of sensors, such as left-right and top-bottom. The ESP32 sends control signals to the horizontal and vertical servo motors based on this comparison. This moves the solar panel so that it is facing the direction with the most light.

4) *Optional IoT Data Transmission:* You can program the ESP32 to send real-time tracking data, like servo angles, LDR readings, or voltage output from the solar panel, to an IoT platform like Blynk or ThingSpeak over Wi-Fi so that you can watch and analyze it from anywhere.

D. Sensor Calibration

To make sure that the tracking system can accurately detect sunlight intensity and move the solar panel correctly, the LDR sensors must be properly calibrated. Calibration helps the ESP32 read light levels correctly and keep its position stable even when the light changes.

1) *LDR Calibration:* Calibration is done to make sure that all LDR sensors give the same output because each one is a little different in how sensitive it is to light. To do this, all of the LDRs are put in front of the same light source, and the ESP32's ADC pins record their analog values. The code uses offset correction factors to make up for any differences between sensors, making sure that all LDRs respond the same way to the same amount of light.

2) *Threshold Adjustment:* Setting a threshold value tells the servo motors when to move. Shadows or reflections can cause small changes in the amount of light that LDRs see. A sensitivity threshold stops the panel from moving too much or too quickly. You can fine-tune the threshold by watching how the servo works in real sunlight.

3) *Servo Position Calibration:* Before they are put into use, the servo motors are calibrated to set their minimum and maximum rotation angles. This makes sure that the solar panel moves smoothly within its physical limits. This stops the tracker structure from rotating too much and putting too much stress on the mechanics.

E. System Assembly and Testing

Before being put outside, the solar tracking system is carefully put together and tested in controlled conditions to make sure it is properly aligned, stable, and works.

1) *Mounting the ESP32 and Sensors:* The solar panel frame has secure mounts for the ESP32 microcontroller, LDR sensors, servo motors, and voltage regulator (LM7806). Four LDR sensors are placed at the corners of the panel, with small barriers between them to make sure they can accurately measure the intensity of directional light. All of the wiring is neatly organized and insulated so that it doesn't short out or hurt the environment.

2) *Mechanical Alignment and Stability Testing:* The servo motor mounts that let the solar panel move up and down and side to side are attached to the panel. The frame is tested to make sure it moves smoothly, which means that the servo motors can handle the weight of the panel without putting too much stress on the machinery. We check the mounting structure to make sure it is balanced and can stand up to things like wind and vibration.

3) *Electrical Testing:* The power supply circuit is checked before it is fully operational. The LM7806 regulator checks the solar panel's output to make sure it provides a steady 6V DC supply to the ESP32 and servos. To make sure that system performance isn't affected, current flow and voltage levels are checked to make sure they don't change.

4) *Initial Tracking Test:* The system is first tested inside with an artificial light source to check the sensor readings and the logic for moving the servos. After confirmation, outdoor tests are done at different times of the day to make sure the solar panel follows the sun's movement correctly. To make sure that tracking is accurate and the system runs smoothly, data like servo angles and light intensity readings are saved.

F. Field Testing and Data Validation

After the solar tracking system is fully put together and calibrated, it is put through a lot of field tests to see how well it tracks, how reliable it is, and how well it works in real-world conditions:

1) *Outdoor Environment Testing:* The solar tracker is set up outside in an open area where it gets natural sunlight all day. The system's ability to follow the sun's path from morning to night is checked. To see how well something works in the real world, we also look at things like temperature, wind, and shade.

2) *Tracking Accuracy Evaluation:* At regular intervals, the solar panel's angle is measured and compared to the theoretical solar position data that was generated using solar position algorithms. To check how accurate the tracking is, any difference between the actual and expected angles is noted.

3) *Battery and Power Consumption Analysis:* It is possible to keep track of the power used by the control system in systems that use microcontrollers or motors to automatically track things. This information helps figure out how much more power is being generated and makes sure that the tracking system uses as little energy as possible.

IV. ADVANTAGES AND DISADVANTAGES

A. Advantages

The proposed *Auto Solar Pilot* system has a number of important benefits that make it an effective and useful way to get the most out of solar energy:

- **Enhanced Energy Efficiency:** The dual-axis tracking mechanism makes sure that the solar panel always faces the sun in the best way possible. This means that it can produce 25–30
- **Low-Cost Implementation:** The system uses cheap and easy-to-find parts like LDR sensors, servo motors, and the ESP32 microcontroller, which makes it affordable for both home and school use.
- **Automation and Reliability:** The ESP32-based control logic moves the panels automatically, so they don't need any human help. This keeps the panels running all the time and cuts down on maintenance needs.
- **Compact and Scalable Design:** The modular design lets the tracker be made bigger for use in factories or smaller for use in portable or educational systems.
- **Sustainability:** The system encourages eco-friendly energy harvesting and lowers the overall carbon footprint by improving solar capture without adding more panels.
- **IoT Integration Capability:** The ESP32's built-in Wi-Fi makes it possible to monitor things from a distance, log data, and connect to solar management platforms that use the Internet of Things (IoT).

B. Disadvantages

Despite its effectiveness, the system has certain limitations that can be addressed in future developments:

- **Dependence on Weather Conditions:** The LDR-based sensing mechanism works best in direct sunlight, but it may not work as well when it's cloudy or dark.
- **Mechanical Wear and Tear:** Servos and joints wear down over time because the panel moves constantly, so they need to be maintained and calibrated mechanically from time to time.
- **Energy Consumption by Tracking Mechanism:** The servos and control electronics use a small amount of the power that is generated, which slightly lowers the overall net gain.
- **Limited Accuracy under Diffuse Illumination:** LDR sensors can give false readings if the lighting is uneven or there are reflections. This can cause the panel to be slightly out of alignment.
- **Environmental Durability:** Prolonged outdoor exposure may affect sensor sensitivity and electronic components if not adequately sealed or protected.

In general, the Auto Solar Pilot works well and is reliable in normal conditions. However, adding hybrid sensor techniques, strong materials, and predictive algorithms could make it even more accurate, efficient, and long-lasting.

V. FINAL ASSESSMENT

We tested and compared the performance of the *Auto Solar Pilot* system to that of regular fixed solar panels to see how well it worked and how practical it was. The results showed that the sunflower-inspired solar tracking mechanism greatly improves system stability, energy efficiency, and accuracy.

The system consistently tracked the sun during daily and seasonal movements, keeping the solar panel almost perpendicular to the sunlight. Experimental comparisons showed that the average power output was much higher than with a fixed panel setup. The servomotors and ESP32 controller used very little energy compared to the energy they gained, which showed that the proposed design was practical.

The project met its goals of making a smart, cheap, and effective solar tracking system. Using LDR sensors, servo control, and the ESP32 microcontroller made sure that the system worked accurately and quickly, with little delay or mechanical jitter. The system is also good for both home and school use of renewable energy because it is modular and can be expanded.

The tracker worked best when the sky was clear, but it didn't work as well when it was cloudy or there wasn't much light, which suggests that hybrid tracking algorithms might be useful in future versions. Using advanced materials and control strategies can also help with mechanical wear and energy optimization.

In conclusion, the Auto Solar Pilot proves that nature-inspired dual-axis solar tracking can improve the collection of renewable energy. Its simplicity, low cost, and ability to be changed make it a good prototype for future growth into IoT-enabled and large-scale solar tracking applications.

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